

OPEN SOURCE · JAVA

Your locator broke. Your test didn't.

A self-healing layer for Playwright and Selenium tests: when the markup changes, it finds the element again, keeps the test running and tells you exactly what to fix.

Playwright

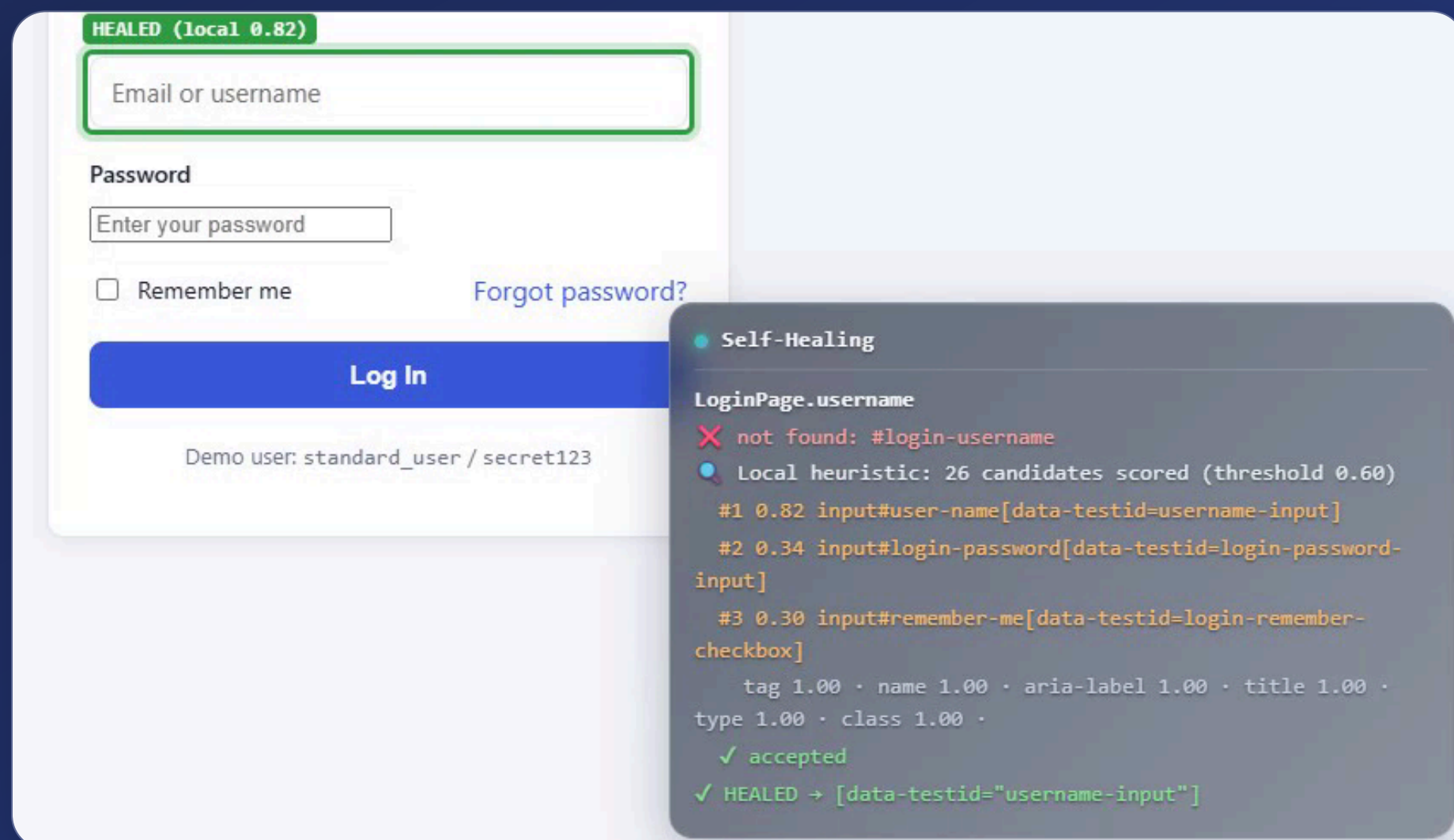
Selenium

JUnit 5 / 4

TestNG

Cucumber

Maven Central



The screenshot shows a login form with the following elements:

- HEALED (local 0.82)**: A green label above the email input field.
- Email or username**: The text inside the email input field.
- Password**: The label for the password input field.
- Enter your password**: The text inside the password input field.
- ☐ Remember me
- [Forgot password?](#)
- Log In**: A blue button.
- Demo user: standard_user / secret123

A self-healing overlay is shown on the right side of the form, displaying the following information:

- Self-Healing**: The title of the overlay.
- LoginPage.username**: The locator being tested.
- not found: #login-username**: The error message.
- Local heuristic: 26 candidates scored (threshold 0.60)**: The heuristic used to find the element.
- #1 0.82 input#user-name[data-testid=username-input]**: The first candidate found.
- #2 0.34 input#login-password[data-testid=login-password-input]**: The second candidate found.
- #3 0.30 input#remember-me[data-testid=login-remember-checkbox]**: The third candidate found.
- tag 1.00 · name 1.00 · aria-label 1.00 · title 1.00 · type 1.00 · class 1.00 ·**: The attributes of the element.
- ✓ accepted**: The result of the heuristic.
- ✓ HEALED → [data-testid="username-input"]**: The final locator used to find the element.

THE PROBLEM

A renamed id should not cost you a red build.

A release renames `#login-username` to `#user-name`.

Nothing is broken for the user - but every test that touches the login fails, and someone spends the morning updating page objects.

✗ Red builds after harmless UI changes

✗ Hours of page-object maintenance after every release

✗ No idea which locators are about to break next

```
<!-- release 2.0 -->
- <input id="login-username" name="username">
+ <input id="user-name" data-testid="username-input">

✗ NoSuchElementException: #login-username
```

Six steps, most of them free

1

Remember

The first green run stores a fingerprint of every element: tag, ids, text, label, position.

2

Detect

A selector finds nothing - the healer takes over instead of failing the test.

3

Match locally

Every element on the page is scored against the fingerprint. Free, milliseconds.

4

Guard

Never another list item, the opposite button, a field for a button, or another locator's element.

5

Ask Claude - optional

Only when the local match is not sure: about \$0.002 per heal.

6

Report

The test passes with a WARN - and a ready-made code fix for the page object.

LIVE, ON A BROKEN DEMO SHOP

26 candidates. One clear winner.

The overlay shows what the healer sees: the old selector is gone, the candidates are scored.

The screenshot shows a login page for 'ShopLab' with the text 'Sign in to your account'. The page contains three input fields: 'Email or username' (scored #1 0.82), 'Enter your password' (scored #2 0.34), and a 'Remember me' checkbox (scored #3 0.30). A 'Log In' button is at the bottom. A 'Demo user: standard_user / secret123' is listed below the button. A 'Forgot password?' link is to the right of the 'Remember me' checkbox. An overlay titled 'Self-Healing' is shown on the right, displaying the following information:

```
Self-Healing

LoginPage.username
✗ not found: #login-username
🔍 Local heuristic: 26 candidates scored (threshold 0.60)
#1 0.82 input#user-name[data-testid=username-input]
#2 0.34 input#login-password[data-testid=login-password-input]
#3 0.30 input#remember-me[data-testid=login-remember-checkbox]
tag 1.00 · name 1.00 · aria-label 1.00 · title 1.00 ·
type 1.00 · class 1.00 ·
✓ accepted
```

HEALED

Milliseconds. \$0.

#product-search → [data-testid="search-box"], local heuristic, confidence 0.69 - the test just goes on.

Products

HEALED (local 0.69)

What are you looking for?

All categories

Recommended

8 products listed

ELECTRONICS

Wireless Headphones

₹1,299.00

Details

Add to Bag

ELECTRONICS

Smart Watch

₹2,499.00

Details

Add to Bag

ELECTRONICS

Mechanical Keyboard

₹1,899.00

Details

Add to Bag

SPORTS

CLOTHING

CLOTHING

Local heuristic: 26 candidates scored (threshold 0.60)

#1 0.69 button#btn-signin[data-testid=signin-button] "Log In"

#2 0.32 form#login-form[data-testid=login-form] "Username Password Remember me Forgot pas..."

#3 0.28 div#login-error[data-testid=login-error-message]

tag 1.00 · data-qa 1.00 · name 1.00 · type 1.00 · class 1.00 · position 0.85 ·

✓ accepted

✓ HEALED → [data-testid="signin-button"]

ProductsPage.search

✗ not found: #product-search

Local heuristic: 99 candidates scored (threshold 0.60)

#1 0.69 input#search-input[data-testid=search-box]

#2 0.34 select#sort-select[data-testid=product-sort-select]

#3 0.29 article#product-card-1[data-testid=product-card-1]

"🎧 ELECTRONICS Wireless Headphones ₹1,299.00"

tag 1.00 · name 1.00 · aria-label 1.00 · title 1.00 · type 1.00 · class 1.00 ·

✓ accepted

✓ HEALED → [data-testid="search-box"]

SAFETY FIRST

A wrong heal is worse than a failed test.

When an element is really gone, the healer says so instead of guessing.

0

wrong elements picked

100%

of deleted elements
correctly refused

```
[FAIL] Form.cancel: '#cancel' could not be healed  
best match: button "Save", score 0.53 < 0.60
```

MEASURED, NOT CLAIMED

Accuracy on a broken shop

LEVEL	LOCAL ONLY (\$0)	+ CLAUDE HAIKU
low ids renamed	100%	100%
medium + names, texts	97.7%	100%
high + moved, re-tagged	87.2%	98.8%
extreme all ids removed	22.1%	97.7%
removed deleted - must not heal	100%	100%
wrong element	0	0

ShopLab demo: 4 pages, 86 elements. Claude cost for all 373 broken elements: \$0.13.

EVERY HEAL IS A TO-DO, NOT A SECRET

The report tells you what to fix

Old → new selector, source line, confidence, cost - and the code fix, as a patch or applied.

Test Automation Report

LanguageEnglishCollapse allSave as PDF

Self-healing locator analysis

Sep 28, 2026, 2:56 PM1 min 32 sMode: AUTOThreshold: 0.6Claude: off

TESTS121 min 32 s total

PASSED CLEAN0no healing needed

PASSED WITH WARN12needs review

FAILED0nothing unhealed

HEALS17local 17 - cache 0 - Claude 0

CLAUDE COST\$0.00000 calls - 0 tokens

Needs review17

These locators were healed during the run. The tests passed, but the page objects should be updated; the suggested fix is on each row.

ELEMENT	OLD → NEW SELECTOR - SUGGESTED FIX	SOURCE	CONFIDENCE	COST
LoginPage.username in 12 test(s)	#login-username → [data-testid="username-input"] src/test/java/com/shoplab/tests/pages/LoginPage.java:16Ready healer.locator("LoginPage.username", "[data-testid="username-input"]")	Local	0.82	\$0.0000
LoginPage.password in 12 test(s)	input[name='password'] → [data-testid="login-password-input"] src/test/java/com/shoplab/tests/pages/LoginPage.java:17Ready healer.locator("LoginPage.password", "[data-testid="login-password-input"]")	Local	0.82	\$0.0000
LoginPage.loginButton in 12 test(s)	[data-testid='login-submit-button'] → [data-testid="signin-button"] src/test/java/com/shoplab/tests/pages/LoginPage.java:18Ready healer.locator("LoginPage.loginButton", "[data-testid="signin-button"]")	Local	0.69	\$0.0000
ProductsPage.addToCart[1]	#add-to-cart-1	Local	0.75	\$0.0000

FITS YOUR STACK

Tested in 20 combinations

Every combination runs the same ten tests - twice: once green, once with a deliberate failure.

	Playwright	Selenium
JUnit 5	✓ OK	✓ OK
JUnit 4	✓ OK	✓ OK
TestNG	✓ OK	✓ OK
Cucumber JUnit 5	✓ OK	✓ OK
Cucumber TestNG	✓ OK	✓ OK
Cucumber JUnit 4	✓ OK	✓ OK

- Parallel runs
- Playwright 1.45 - 1.63
- Selenium 4.21 - 4.49
- Java 17 / 21 / 25
- Linux & Windows
- Maven & Gradle

GET STARTED

One dependency. One settings file.

Free, open source, on Maven Central.

```
<!-- pom.xml -->
<dependency>
  <groupId>io.github.yasindeger48</groupId>
  <!-- or healer-selenium -->
  <artifactId>healer-playwright</artifactId>
  <version>2.2.0</version>
</dependency>

# src/test/resources/healer.properties
browser.name=msedge
browser.headless=true
healer.mode=auto           # auto | suggest | off
healer.llm.enabled=false   # Claude, optional
healer.report.open=onWarn
```

github.com/YasinDeger48/selfhealing-framework